

# DEVA ANAND

devaanand@umass.edu • (413) 315-1715 • devaanand.com • linkedin.com/in/devaaa • github.com/Deva-1903

## Education

|                                                                                                                       |           |
|-----------------------------------------------------------------------------------------------------------------------|-----------|
| <b>MS, Computer Science</b> , University of Massachusetts Amherst                                                     | 2025–2027 |
| <i>Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering</i> |           |
| <b>BE, Computer Science and Engineering</b> , Vels Institute of Science, Technology and Advanced Studies              | 2019–2023 |

## Professional Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Wysa</b> , Backend Engineer   <i>Bengaluru, India</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Jul 2023 – Jul 2025 |
| <ul style="list-style-type: none"><li>Engineered a well-architected multi-tenant NHS eTriage backend (Node.js, MongoDB) for <b>8+ UK clinical clients</b>, processing <b>10,000+ monthly triage submissions</b> via Mayden iaptus REST integrations; contributed to <b>\$340K+</b> in revenue.</li><li>Owned backend reliability and quality: added integration tests for regression protection, monitored production logs and debugged tenant-specific incidents, hardened auth and rate limiting, and remediated <b>20+ VAPT findings</b>.</li><li>Architected a full-stack internal platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by <b>100+ hours/month</b> and lifting throughput by <b>60%</b>.</li><li>Served as primary technical POC, collaborating cross-functionally with clinical and product teams, onboarded a second developer to on-call readiness in <b>~6 weeks</b>, and piloted a COO-sponsored GitHub Copilot experiment.</li></ul> |                     |
| <b>Cario Growth Services</b> , Backend Developer Intern   <i>Chennai, India</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Nov 2022 – May 2023 |
| <ul style="list-style-type: none"><li>Designed and shipped <b>20+ RESTful APIs</b> (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.</li><li>Migrated core repositories from JavaScript to TypeScript and shipped a production LLM auto-commenting feature (<b>4-bit quantized Llama-7B via llama.cpp in C/C++</b>) with async queued generation and content-safety filtering.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |
| <b>Freelance / Open Source</b> , Full-Stack Engineer   <i>Remote</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Jul 2022 – Aug 2023 |
| <ul style="list-style-type: none"><li>Built and deployed <b>Underdogs Fitness</b> (Website), a production MERN gym-management platform with Stripe + manual-cash hybrid payments, cron automation, and multi-branch support; live <b>3+ years</b>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |

## Projects

|                                                                                                                                                                                                                                                                                                      |                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Cache- and SIMD-Aware Matrix Multiplication</b> (GitHub)                                                                                                                                                                                                                                          | C++, AVX/SIMD, OpenMP, Linux perf       |
| <ul style="list-style-type: none"><li>Optimized C++ matrix multiplication from a naive baseline through cache-aware tiling, <b>AVX/SIMD</b> vectorization, register-aware micro-kernels, and <b>OpenMP</b>, profiling L1/L3/TLB misses with <b>perf</b> and inspecting generated assembly.</li></ul> |                                         |
| <b>Spark ETL Performance Optimization</b> (GitHub)                                                                                                                                                                                                                                                   | PySpark, Spark UI, SQL, NYC TLC Dataset |
| <ul style="list-style-type: none"><li>Cut end-to-end Spark ETL runtime <b>~25%</b> (17–19 min to 14.3 min) on a <b>~1.4B-row</b> dataset via broadcast joins, AQE, and skew-join handling, cutting the tail-latency ratio <b>2.14x to 1.74x</b> (profiled via Spark UI).</li></ul>                   |                                         |
| <b>AgenticSearch – Provenance-First Entity Discovery</b> (GitHub   Demo)                                                                                                                                                                                                                             | Python, FastAPI, OpenAI / Groq, SQLite  |
| <ul style="list-style-type: none"><li>Built a well-architected multi-stage backend service (Python, FastAPI) turning open-ended queries into structured entity tables with deterministic extraction and <b>cell-level provenance</b>; hardened with <b>150+ tests</b>.</li></ul>                     |                                         |

## Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

## Skills

**Languages & Databases:** C++, Java, Python, TypeScript, JavaScript, SQL, Bash, PostgreSQL, MongoDB, SQLite  
**Frameworks / Tools:** Node.js, Express, FastAPI, React, Next.js, Git, GitHub Actions, REST APIs  
**Algorithms / Systems:** data structures & algorithms, AVX/SIMD, cache-aware design, OpenMP, Spark / PySpark, perf profiling, Linux  
**ML & AI:** PyTorch, Hugging Face Transformers, LLM integration, RAG, NumPy, evaluation  
**Cloud / Platforms:** AWS, GCP, Azure, Docker, MongoDB Atlas

## Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.